

Tools workshop

The 19th and 20th centuries saw the development of a wide variety of tools that help us work more quickly, accurately, and efficiently. These tools have revolutionized home improvements, allowing people to measure things precisely and fasten them together securely. In industry, the use of computer control and lasers paved the way for new and improved cutting tools.



Pocket tape measure

Tape measure

- **What?** Retractable pocket tape measure
- **Who?** William H. Bangs
- **Where and when?** US, 1864

Retractable tape measures can fit into a pocket or toolkit, but contain enough tape to measure distances of several yards.

Typical tape measures have 25 ft (7.6 m) of tape. A locking button allows a length of tape to be held in position out of its case. Sliding the button makes a spring pull the tape back into its case and wind it around a coil so it can be stored conveniently.

The measurement on the ruler is shown on an electronic display.



Digital caliper

Micrometer

- **What?** Micrometer screw gauge
- **Who?** Jean Palmier
- **Where and when?** France, 1848

A caliper measures the distance between opposite sides of an object. In 1848, Jean Palmier received a patent for the micrometer screw gauge, a type of caliper that uses a screw to accurately measure very small objects. The object is placed between a fixed frame and a screw. Measuring the rotation of the screw as it moves toward the object shows how far it has moved forward much more precisely. Modern-day calipers show distances on a digital display.

Electric cordless hand drill



The drill bit rotates to cut a hole in a surface.

The trigger allows the user to start and stop the drill.

Electric drill

- **What?** Drill driven by an electric motor
- **Who?** Arthur James Arnot and William Blanch Brain
- **Where and when?** Australia, 1889

Arthur James Arnot and William Blanch Brain invented the first drill to be powered by an electric motor, which was faster and more efficient than existing drills. This first electric drill wasn't portable, but the portable handheld drill was invented just six years later, in 1895, by German brothers Wilhelm and Carl Fein.